SQL Clauses

Nicole Carlet

LEARNING OBJECTIVES

After completing this topic, the students should be able to demonstrate basic skill in SQL such as the use of different Clauses, Data Definition Language and Data Manipulation Language.

STRUCTURED QUERY LANGUAGE (SQL)

- The SQL language may be considered one of the major reasons for the commercial success of relational databases because it became a standard for relational databases.
- Originally, SQL was called SEQUEL (Structured English QUEry Language) and was designed and implemented at IBM Research as the interface for an experimental relational database system called SYSTEM R.
- SQL is now the standard language for commercial relational DBMSs.

STRUCTURED QUERY LANGUAGE (SQL)

- The SQL language is sub-divided into several language elements such as **Clauses** which are in some cases optional and are basic components of statements, **Expression** which is a combination of one or more values, operators and SQL functions that evaluate to a value, and **Predicate** which is a condition expression that evaluates to a Boolean value (true or false). Likewise, predicate specifies conditions that can be evaluated which are used to limit the effects of statements and queries, or to change program flow. Usually, predicate uses comparison operators such as =, <>, <, >, <=, >= BETWEEN, and LIKE.

SQL CLAUSES

As mentioned above that Clauses are in some cases optional and are basic components of statements. This time, let's describe each clause and use some examples. We will start from the basic queries until we have complex queries.

SQL CLAUSES

SELECT, FROM, and WHERE Clause

According to Elmasri, et al. (2011), the following is the basic form for SELECT statement which is composed of three clauses (SELECT, FROM, and WHERE):

```
SELECT   <attribute list>
FROM     <table list>
WHERE    <condition>;
```

SQL CLAUSES

where

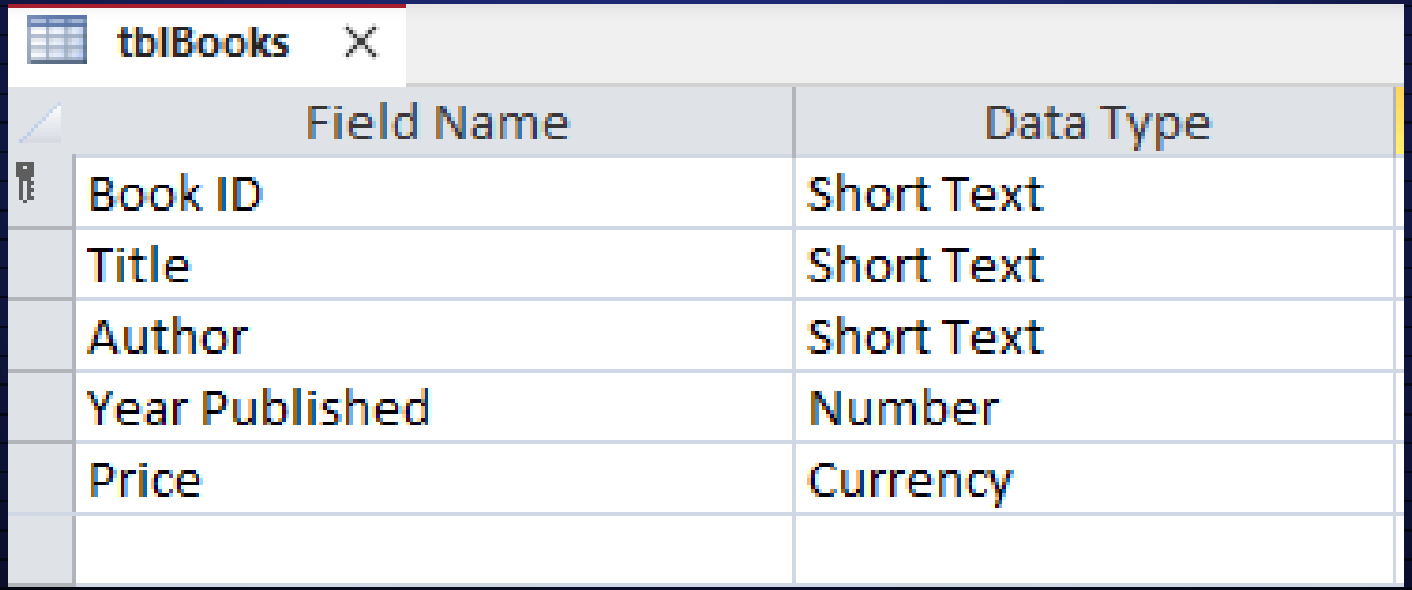
<attribute list> is a list of attribute names whose values are to be retrieved by the query.
<table list> is a list of the relation names required to process the query.
<condition> is a conditional (Boolean) expression that identifies the tuples to be retrieved by the query.

SQL CLAUSES

In addition, the SELECT clause lists the attributes to be retrieved, and the FROM clause specifies all relations (tables) needed in the simple query. The WHERE clause identifies the conditions for selecting the tuples from these relations, including join conditions if needed (Elmasri, et al., 2011).

SQL CLAUSES

Now, let us try to use the basic SQL form using Microsoft Access (MS Access). You need to create a table for books (tblBooks) using the table structure below.



Field Name	Data Type
Book ID	Short Text
Title	Short Text
Author	Short Text
Year Published	Number
Price	Currency

SQL CLAUSES

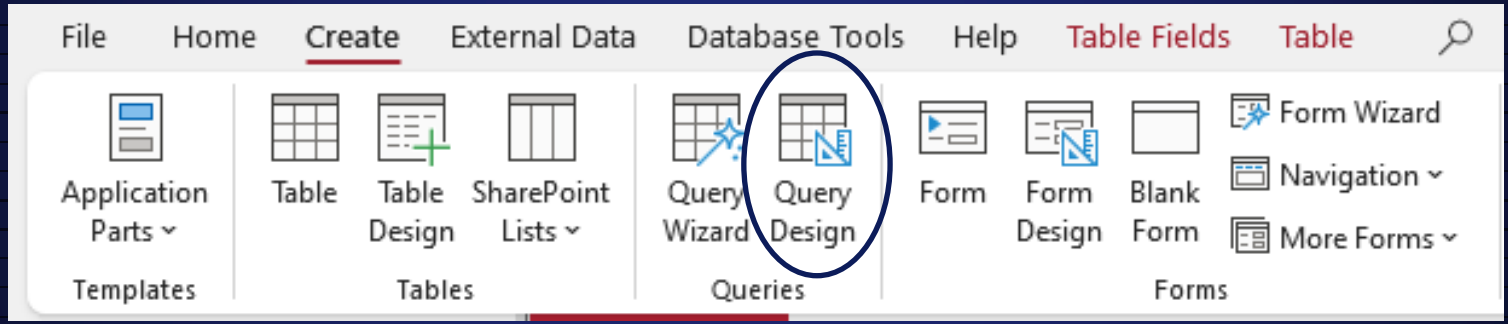
Let us also add at least five (5) records in our newly created table. You may use the same records below or you may want to insert your preferred records.

tblBooks

Book ID	Title	Author	Year Publish	Price
CB-001	Fundamentals of DBMS	Elmarie Suetos	2015	₱4,500.00
CB-002	MS Office Tutorial	Jake Maggay	2019	₱2,500.00
CB-003	SQL Basics	Manny Alipio	2018	₱2,500.00
CB-004	Introduction to Database	Michelle Alar	2016	₱2,000.00
CB-005	SQL: Concepts and Applications	Vicky Agcaoili	2020	₱2,000.00
			0	₱0.00

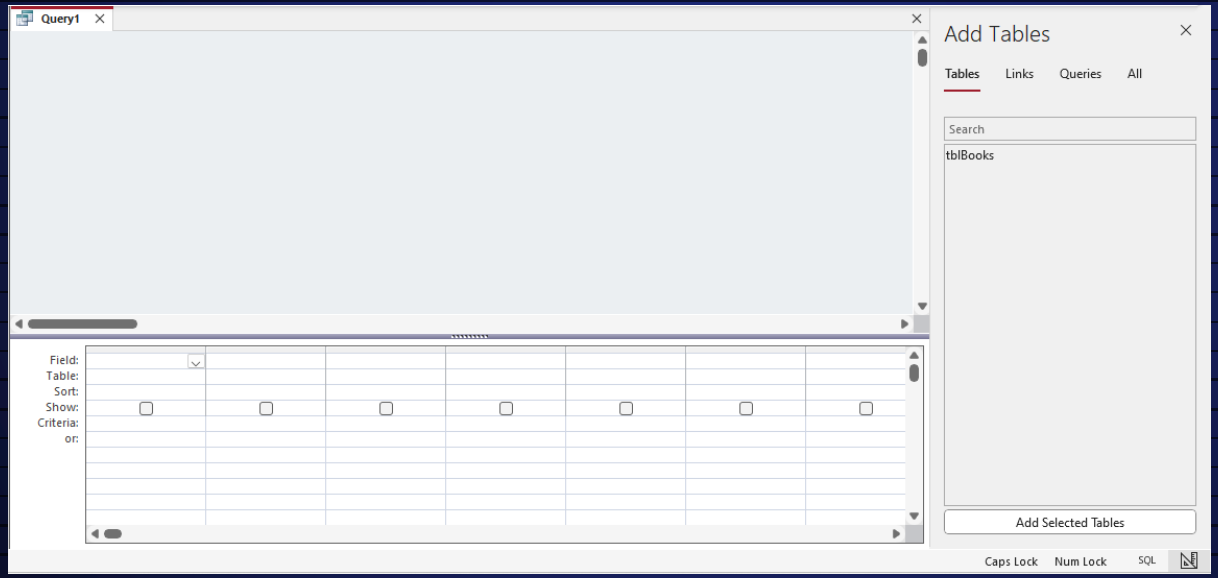
SQL CLAUSES

Using a query, let's try to retrieve records applying the basic SQL form. On the CREATE table of your MS Access application, click Query Design menu (see figure below).



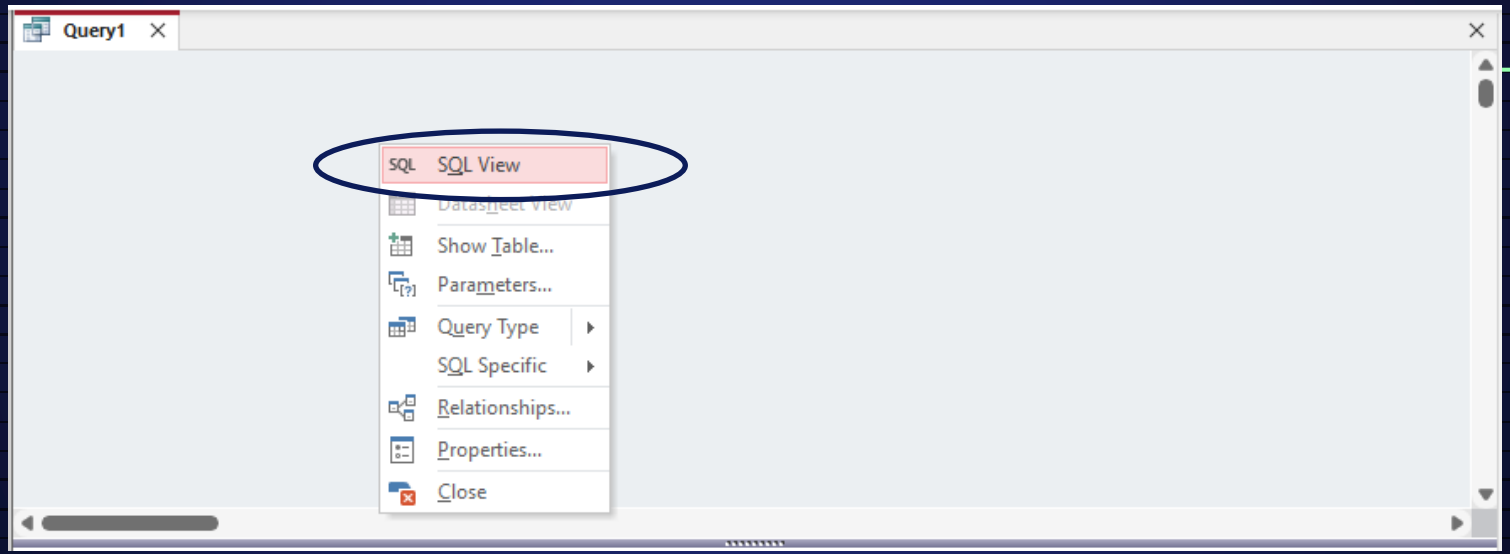
SQL CLAUSES

As you click Query Design button, a Show Table dialog appears displaying the tables in your database. Just close the dialog box because we do not need to select any table from the list.



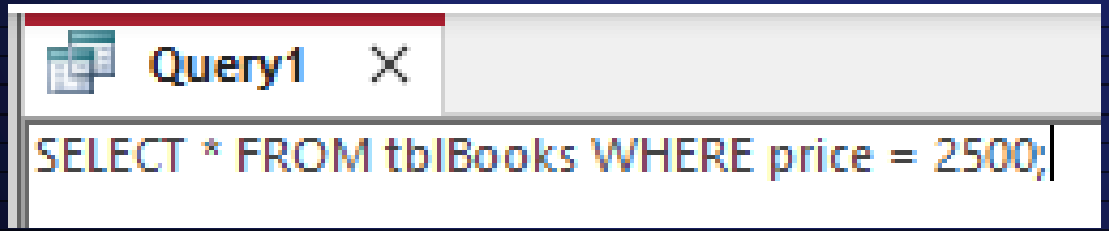
SQL CLAUSES

Right Click and click SQL View.



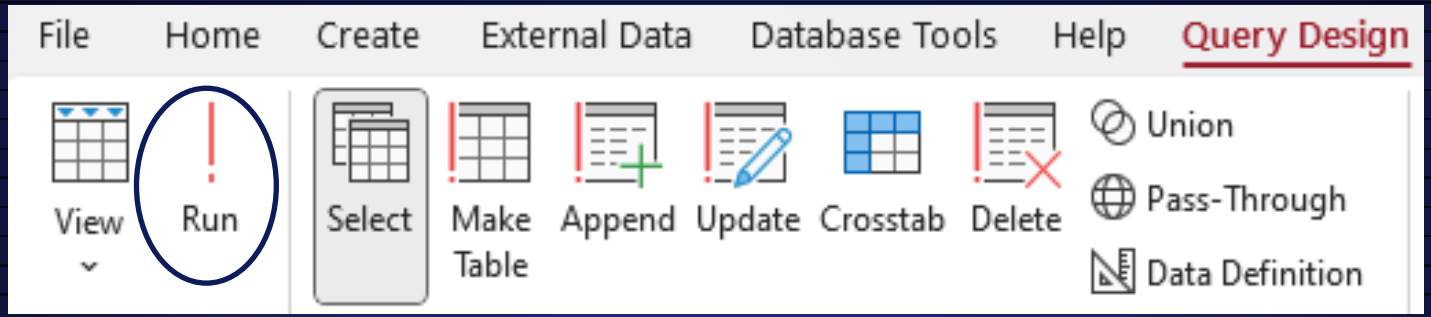
SQL CLAUSES

We are now ready to write our first basic SQL statement. Let us retrieve all books which price is 2,500 using the statement below. Always remember that unlike string values, numeric values do not need to be enclosed with a pair of single quote (‘ ’). Also, SQL statements are case insensitive which means that you can use either upper case and lower case letters or combination of upper case and lower case letters; however, for readability’s sake of this tutorial, we shall use upper case letters for clauses.



SQL CLAUSES

Let us run the SQL statement above by clicking the Run button under Query Design Tab as shown in the figure below.



SQL CLAUSES

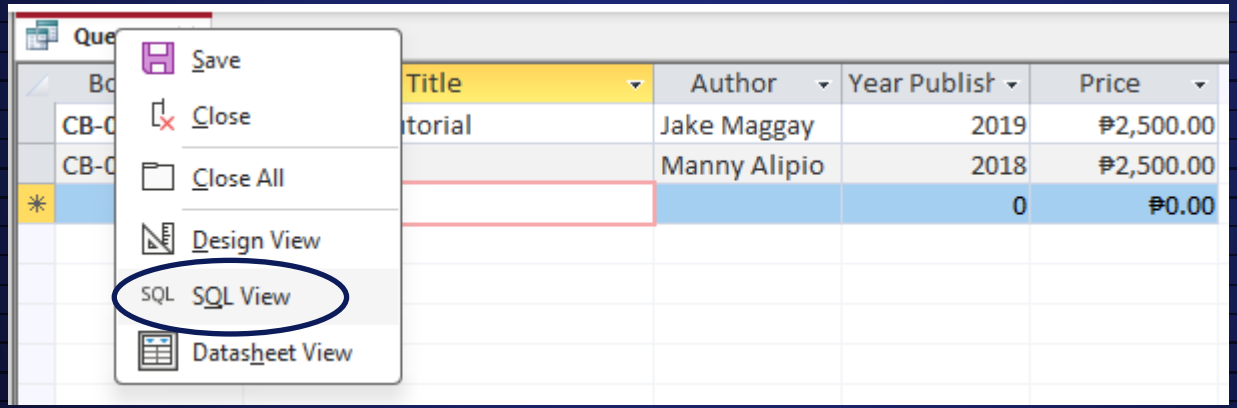
As you have observed in the result, there are only two records displayed by the query. In addition, only the books having 2,500 as the price were retrieved as the price matched with our condition (Price = 2500).

Query1

Book ID	Title	Author	Year Publisł	Price
CB-002	MS Office Tutorial	Jake Maggay	2019	₱2,500.00
CB-003	SQL Basics	Manny Alipio	2018	₱2,500.00
*			0	₱0.00

SQL CLAUSES

Let us try another SQL statement. This time, we will select a book based on its title. But before that, we shall return or query to SQL View. Right Click on the query name and click SQL View.



SQL CLAUSES

Here is another SQL statement which uses a string value as a condition in retrieving a particular record. You have also observed that the string value is enclosed with a pair of single quote ('sql basics').

Query1

```
SELECT * FROM tblBooks WHERE title = 'sql basics';
```

Query1

Book ID	Title	Author	Year Publish	Price
CB-003	SQL Basics	Manny Alipio	2018	₱2,500.00
*			0	₱0.00

SQL CLAUSES

Based on the two examples, you have observed that we used asterisk (*) on the attribute list of our statements which indicates or represents that we are displaying all columns. The next examples will demonstrate how to retrieve specific columns.

SQL CLAUSES

We will use the previous examples; however, we will try to specify the desired columns to be displayed by the query. The SQL statement below will only display the columns *Title*, *Author* and *Price*.

Query1

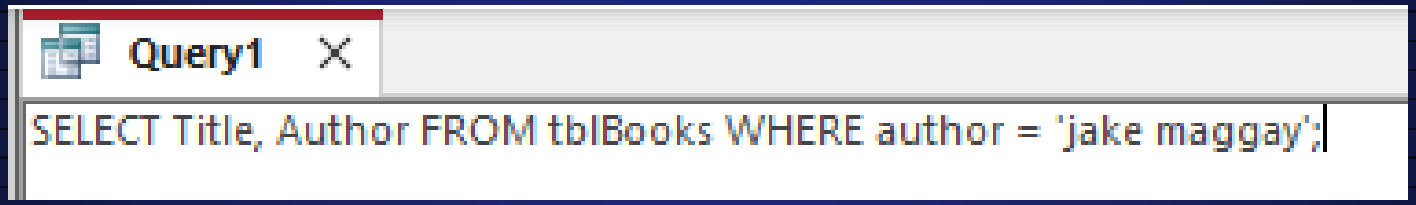
```
SELECT Title, Author, Price FROM tblBooks WHERE price = 2500;
```

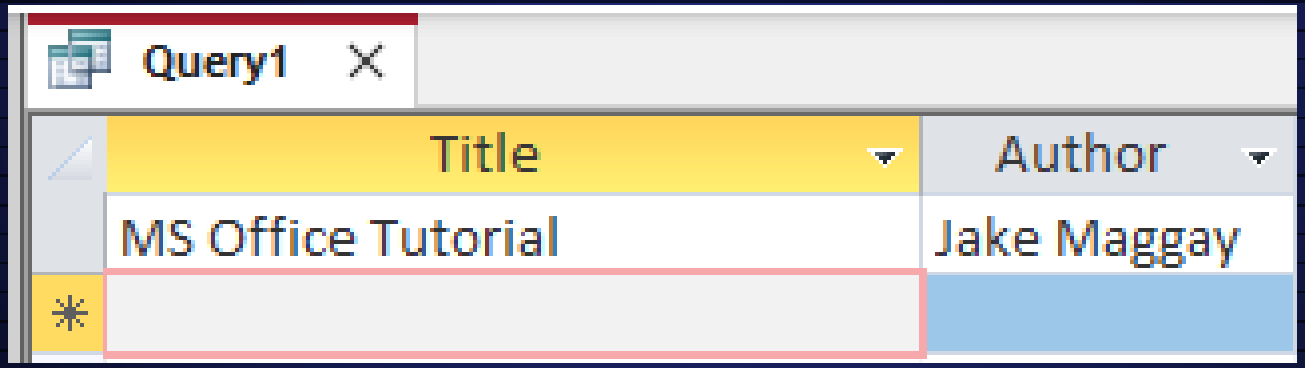
Query1

Title	Author	Price
MS Office Tutorial	Jake Maggay	₱2,500.00
SQL Basics	Manny Alipio	₱2,500.00
*		₱0.00

SQL CLAUSES

Let's have another example where the column Author will serve as the criteria in searching specific record as shown below.





SQL CLAUSES

If you want to retrieve all records without condition, you just have to exclude the WHERE clause in the SQL statement. The sample SQL statement and sample result of the query is shown below.



Query1

Title	Author
Fundamentals of DBMS	Elmarie Suetos
MS Office Tutorial	Jake Maggay
SQL Basics	Manny Alipio
Introduction to Database	Michelle Alar
SQL: Concepts and Applications	Vicky Agcaoili

ORDER BY, GROUP BY, and HAVING Clause

This time, we will also learn other clauses such as ORDER BY, GROUP BY, and HAVING clause. In addition, these clauses are optional. You may or may not include them in your statement depending on the result you want to retrieve and display.

ORDER BY, GROUP BY, and HAVING Clause

Elmasri, et al. (2011) said that ORDER BY clause specifies an order for displaying the results of a query. In addition, it is used to sort the selected records in either ascending (ASC) or descending (DESC) order based on the specified column or columns.

ORDER BY, GROUP BY, and HAVING Clause

Query1

```
SELECT Title, Author FROM tblBooks ORDER BY title ASC;
```

Query1

Title	Author
Fundamentals of DBMS	Elmarie Suetos
Introduction to Database	Michelle Alar
MS Office Tutorial	Jake Maggay
SQL Basics	Manny Alipio
SQL: Concepts and Applications	Vicky Agcaoili

ORDER BY, GROUP BY, and HAVING Clause

Query1

```
SELECT Title, Author FROM tblBooks ORDER BY title DESC;
```

Query1

Title	Author
SQL: Concepts and Applications	Vicky Agcaoili
SQL Basics	Manny Alipio
MS Office Tutorial	Jake Maggay
Introduction to Database	Michelle Alar
Fundamentals of DBMS	Elmarie Suetos

ORDER BY, GROUP BY, and HAVING Clause

Query1

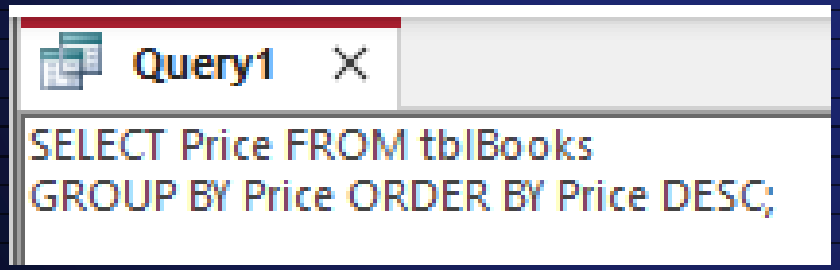
```
SELECT BookID, Title, Price, Price*0.06 AS Sales_Tax
FROM tblBooks
WHERE Price > 2000
ORDER BY Price DESC;
```

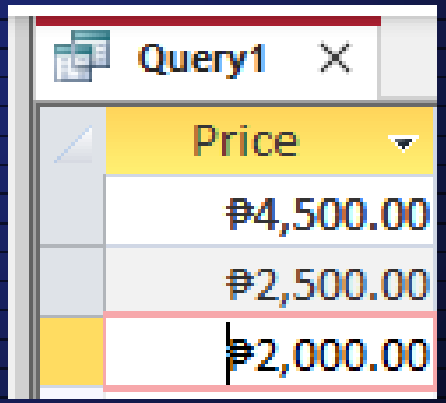
Query1

BookID	Title	Price	Sales_Tax
CB-001	Fundamentals of DBMS	₱4,500.00	270
CB-003	SQL Basics	₱2,500.00	150
CB-002	MS Office Tutorial	₱2,500.00	150
*		₱0.00	

ORDER BY, GROUP BY, and HAVING Clause

This time, we will also learn the use of GROUP BY clause. According to Elmasri, et al. (2011), GROUP BY clause specifies the grouping attributes, which should also appear in the SELECT clause. The GROUP BY clause will only group records and columns having exactly the same value. Below is an example on how to use GROUP BY clause with its sample output.

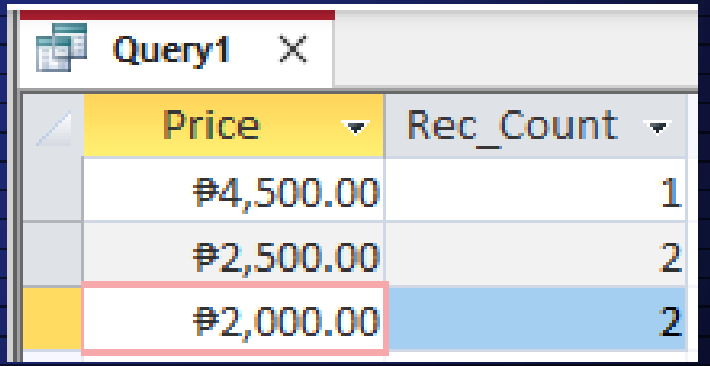




ORDER BY, GROUP BY, and HAVING Clause

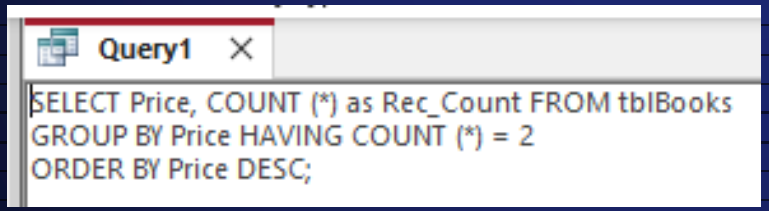
We can also count how many record were grouped by adding the COUNT (*) function in our SQL statement. The COUNT function simply counts the number of rows or records.

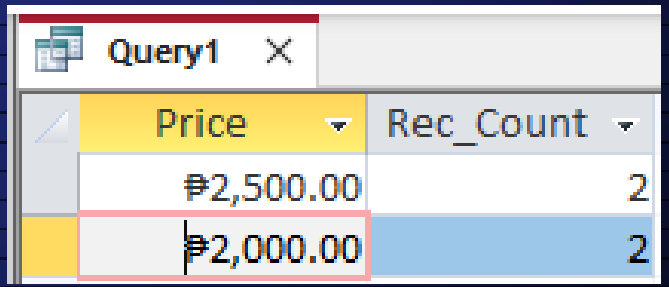




ORDER BY, GROUP BY, and HAVING Clause

The last clause in this topic is the HAVING clause. SQL provides a HAVING clause, which can appear in conjunction with a GROUP BY clause. HAVING provides a condition on the summary information regarding the group of tuples associated with each value of the grouping attributes. Only the groups that satisfy the condition are retrieved in the result of the query (Elmasri, et al., 2011).





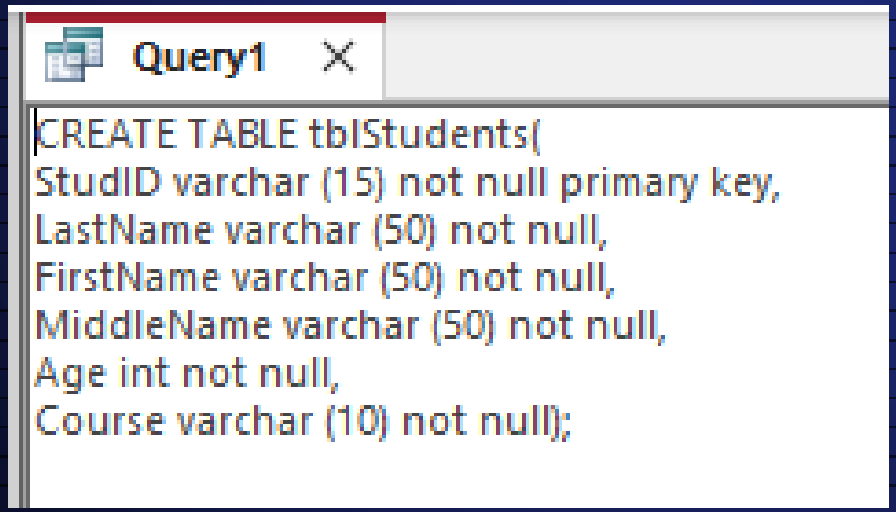
Data Definition Language (DDL)

Data definition language (DDL), is used by the DBA and by database designers to define both schemas. The DBMS will have a DDL compiler whose function is to process DDL statements in order to identify descriptions of the schema constructs and to store the schema description in the DBMS catalog (Elmasri, et al., 2011). In simpler description, DDL is use to define the structure of the table. In this sub-topic, we will discuss and demonstrate the use of DDL such as CREATE, ALTER, DROP and TRUNCATE.

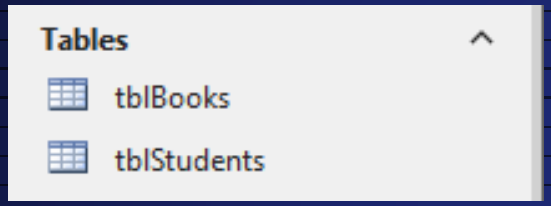
Data Definition Language (DDL)

CREATE Command

The main SQL command for data definition is the CREATE statement, which can be used to create schemas and tables.



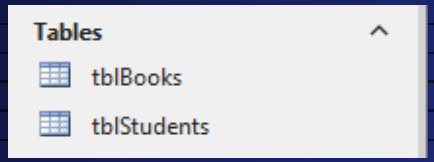
Data Definition Language (DDL)

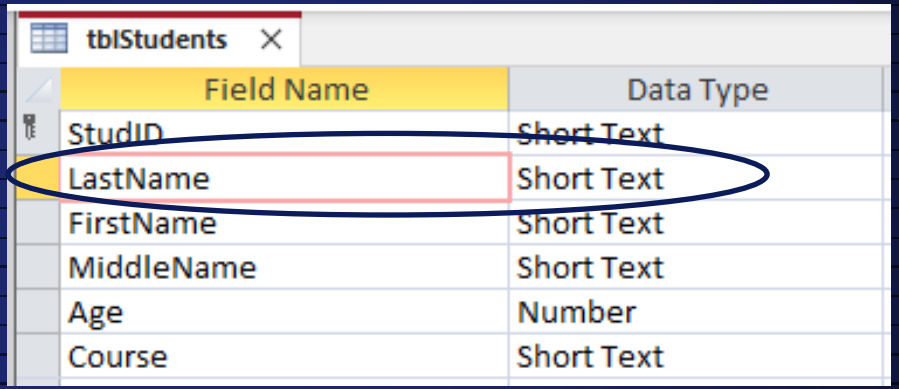


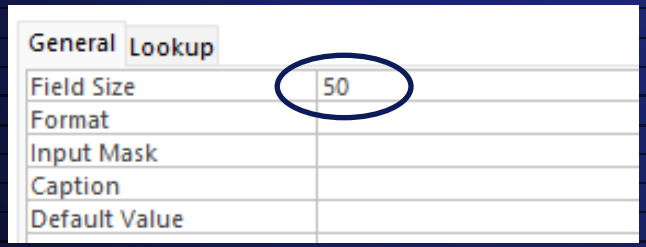
tblStudents

Field Name	Data Type
StudID	Short Text
LastName	Short Text
FirstName	Short Text
MiddleName	Short Text
Age	Number
Course	Short Text

Data Definition Language (DDL)



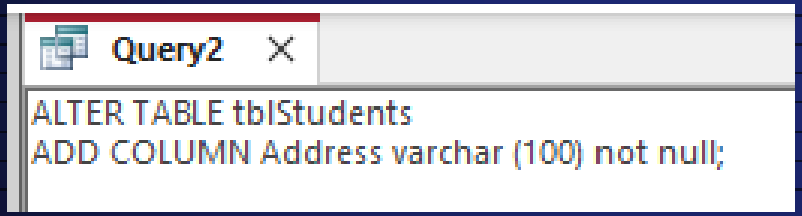




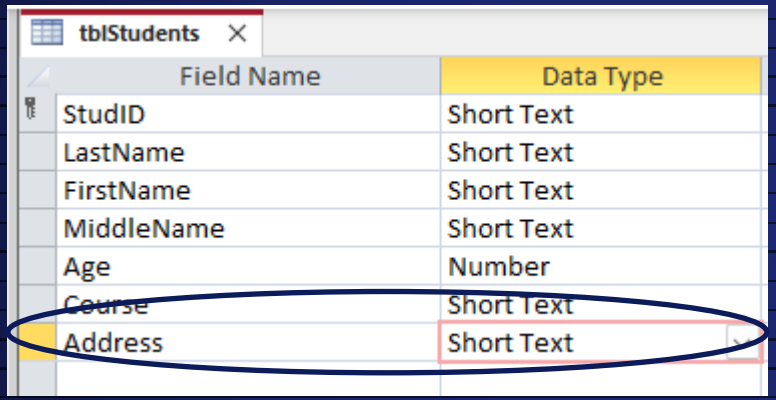
Data Definition Language (DDL)

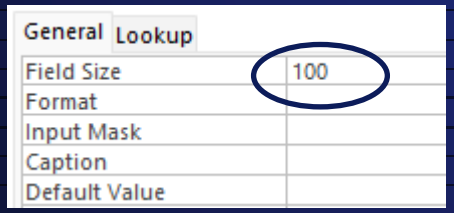
ALTER Command

The next command is the ALTER command which is use to change the definition or structure of tables such as adding and dropping of columns (attributes), changing column name or adding and dropping of table constraints.



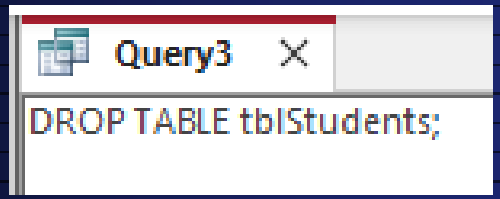
Data Definition Language (DDL)

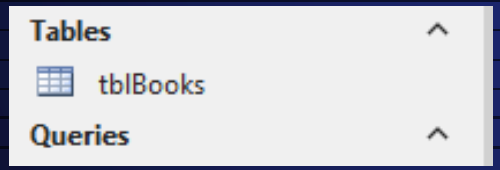




Data Definition Language (DDL)

The next command is the DROP command which can be used to drop named schema elements, such as tables.





Data Manipulation Language (DML)

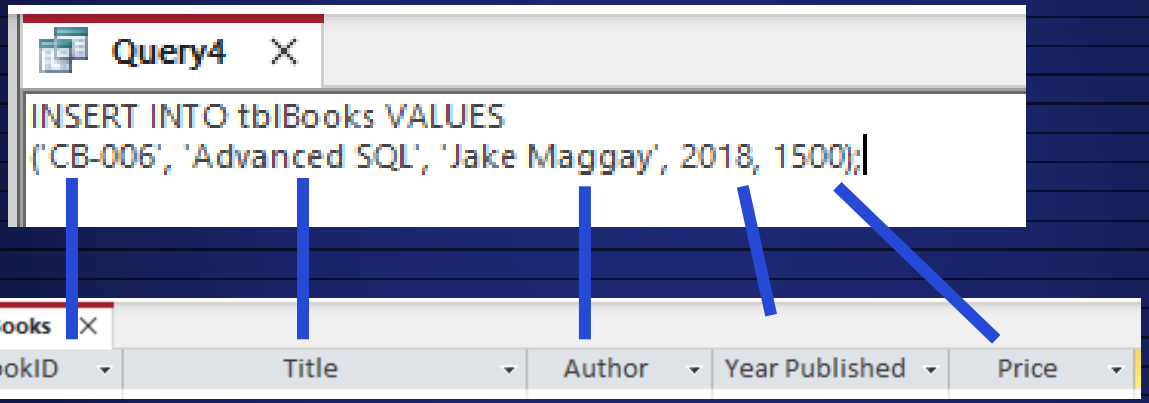
Elmasri, et al. (2011) described DML as a set of operations or language used to manipulate the database. Typical manipulations include retrieval (SELECT), insertion (INSERT), deletion (DELETE), and modification (UPDATE) of the data.

Data Manipulation Language (DML)

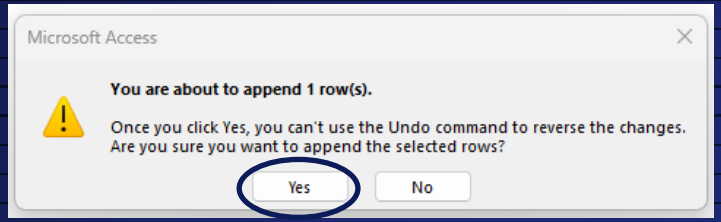
INSERT Command

Now, let us discuss and use the INSERT command. In its simplest form, INSERT is used to add a single tuple (row or record) to a relation (table). We must specify the relation name and a list of values for the tuple. The values should be listed in the same order in which the corresponding attributes were specified in the CREATE TABLE command. For example, to add a new tuple to tblBooks table, we need to insert five (5) values because there are five (5) attributes or columns in our table.

Data Manipulation Language (DML)



Data Manipulation Language (DML)

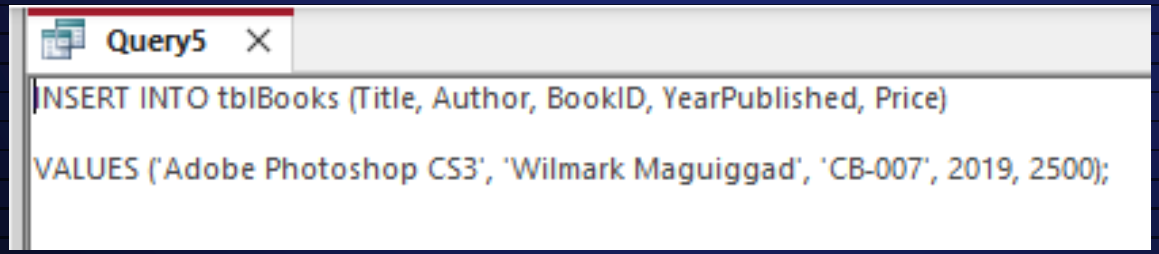


tblBooks

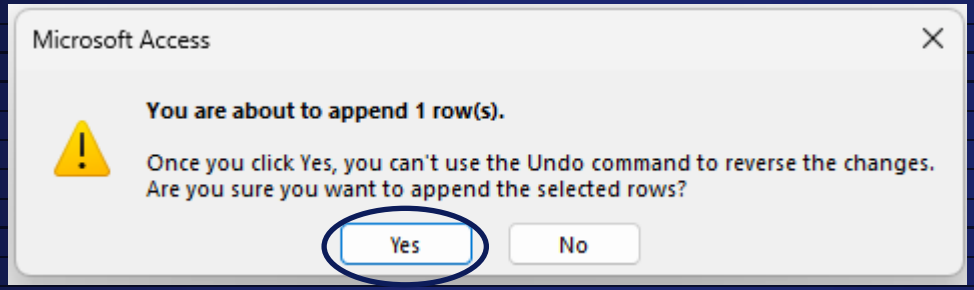
BookID	Title	Author	Year Publish	Price
CB-001	Fundamentals of DBMS	Elmarie Suetos	2015	₱4,500.00
CB-002	MS Office Tutorial	Jake Maggay	2019	₱2,500.00
CB-003	SQL Basics	Manny Alipio	2018	₱2,500.00
CB-004	Introduction to Database	Michelle Alar	2016	₱2,000.00
CB-005	SQL: Concepts and Applications	Vicky Agcaoili	2020	₱2,000.00
CB-006	Advanced SQL	Jake Maggay	2018	₱1,500.00
			0	₱0.00

Data Manipulation Language (DML)

Another form of the INSERT statement allows the user to specify explicit attribute names that correspond to the values provided in the INSERT command. This is useful if a relation has many attributes but only a few of those attributes are assigned values in the new tuple. In the example below, the values are not in the same order with that of the columns; however, we specified the order of the columns in the statement (Title, Author, BookID, YearPublished, Price).



Data Manipulation Language (DML)



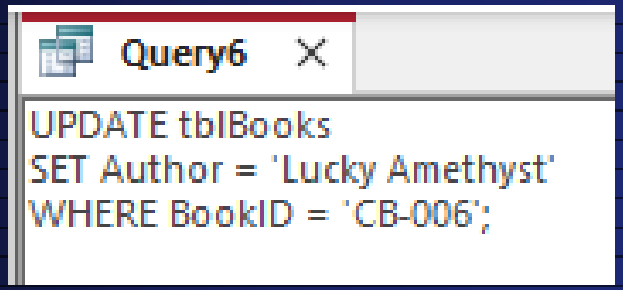
tblBooks

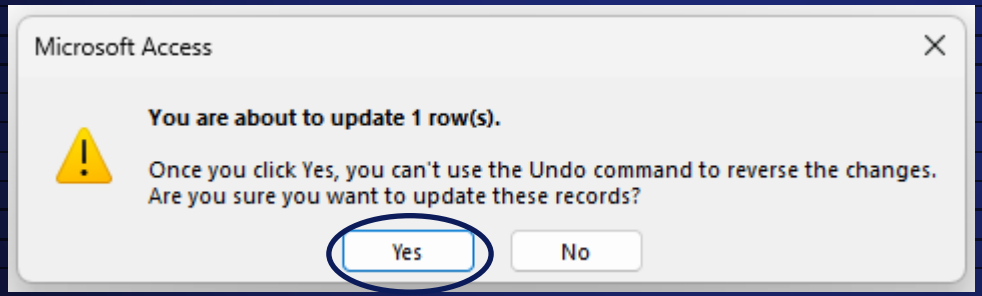
BookID	Title	Author	YearPublish	Price	Click to Add
CB-001	Fundamentals of DBMS	Elmarie Suetos	2015	₱4,500.00	
CB-002	MS Office Tutorial	Jake Maggay	2019	₱2,500.00	
CB-003	SQL Basics	Manny Alipio	2018	₱2,500.00	
CB-004	Introduction to Database	Michelle Alar	2016	₱2,000.00	
CB-005	SQL: Concepts and Applications	Vicky Agcaoili	2020	₱2,000.00	
CB-006	Advanced SQL	Jake Maggay	2018	₱1,500.00	
CB-007	Adobe Photoshop CS3	Wilmark Maguiggad	2019	₱2,500.00	
*			0	₱0.00	

Data Manipulation Language (DML)

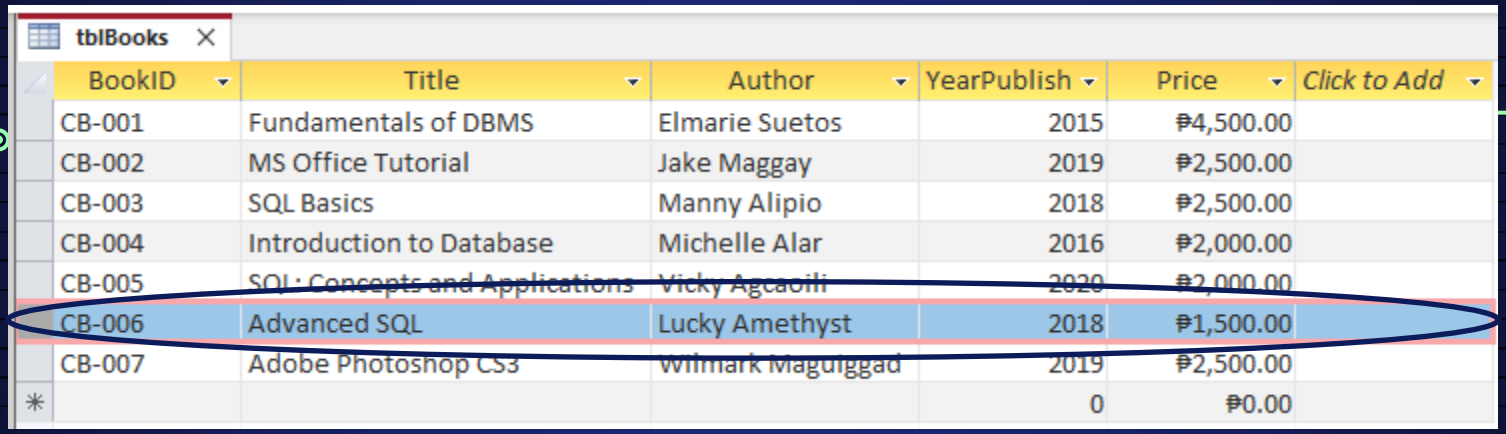
If there are existing records and you need to modify for some reasons, you can always do that using the UPDATE command. The UPDATE command is used to modify attribute values of one or more selected tuples. A WHERE clause in the UPDATE command selects the tuples to be modified from a single relation. An additional SET clause in the UPDATE command specifies the attributes to be modified and their new values.

Data Manipulation Language (DML)





Data Manipulation Language (DML)



tblBooks

BookID	Title	Author	YearPublish	Price	Click to Add
CB-001	Fundamentals of DBMS	Elmarie Suetos	2015	₱4,500.00	
CB-002	MS Office Tutorial	Jake Maggay	2019	₱2,500.00	
CB-003	SQL Basics	Manny Alipio	2018	₱2,500.00	
CB-004	Introduction to Database	Michelle Alar	2016	₱2,000.00	
CB-005	SQL: Concepts and Applications	Vicky Agcaoili	2020	₱2,000.00	
CB-006	Advanced SQL	Lucky Amethyst	2018	₱1,500.00	
CB-007	Adobe Photoshop CS3	Wilmark Maguiggad	2019	₱2,500.00	
*			0	₱0.00	

Data Manipulation Language (DML)

You can also modify multiple or all columns in a single UPDATE command by using a comma (,) to separate each column.

Query7

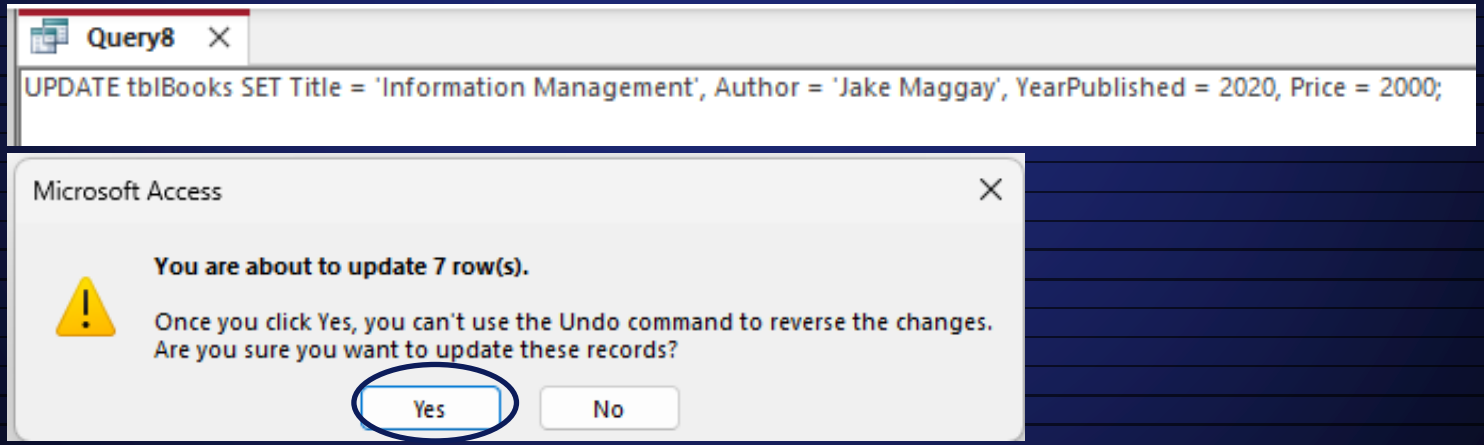
```
UPDATE tblBooks SET YearPublished = 2020, Price = 2000
WHERE BookID = 'CB-006';
```

tblBooks

BookID	Title	Author	YearPublish	Price
CB-001	Fundamentals of DBMS	Elmarie Suetos	2015	₱4,500.00
CB-002	MS Office Tutorial	Jake Maggay	2019	₱2,500.00
CB-003	SQL Basics	Manny Alipio	2018	₱2,500.00
CB-004	Introduction to Database	Michelle Alar	2016	₱2,000.00
CB-005	SQL: Concepts and Applications	Vicky Agcoili	2020	₱2,000.00
CB-006	Advanced SQL	Lucky Amethyst	2020	₱2,000.00
CB-007	Adobe Photoshop CS3	Wilmark Maguiggad	2019	₱2,500.00

Data Manipulation Language (DML)

It is highly recommended that a WHERE clause should be included in the UPDATE command to select a specific record. Without the WHERE clause, all records will be modified or updated. Unfortunately, changes cannot be undone.



Data Manipulation Language (DML)

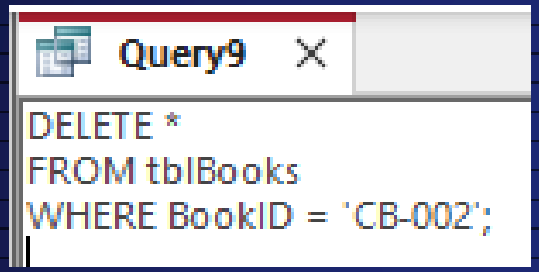
tblBooks

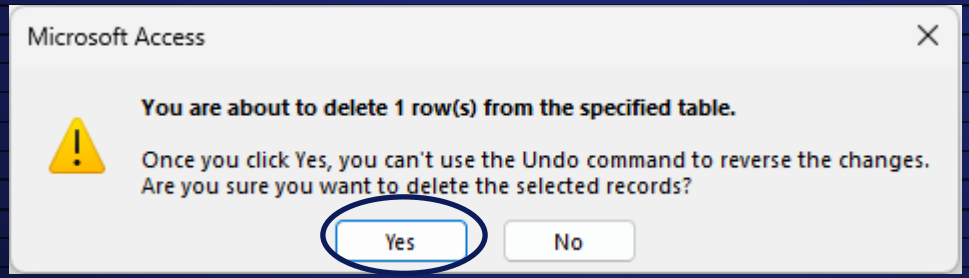
BookID	Title	Author	YearPublish	Price
CB-001	Information Management	Jake Maggay	2020	₱2,000.00
CB-002	Information Management	Jake Maggay	2020	₱2,000.00
CB-003	Information Management	Jake Maggay	2020	₱2,000.00
CB-004	Information Management	Jake Maggay	2020	₱2,000.00
CB-005	Information Management	Jake Maggay	2020	₱2,000.00
CB-006	Information Management	Jake Maggay	2020	₱2,000.00
CB-007	Information Management	Jake Maggay	2020	₱2,000.00

Data Manipulation Language (DML)

If have entered a wrong data, you can delete it by using the DELETE command. The DELETE command removes tuples (records) from a relation. It includes a WHERE clause, similar to that used in an SQL query, to select the tuples to be deleted. Tuples are explicitly deleted from only one table at a time. A missing WHERE clause specifies that all tuples in the relation are to be deleted; however, the table remains in the database as an empty table (Elmasri, et al., 2011).

Data Manipulation Language (DML)





Data Manipulation Language (DML)

tblBooks

BookID	Title	Author	YearPublish	Price
CB-001	Information Management	Jake Maggay	2020	₱2,000.00
CB-003	Information Management	Jake Maggay	2020	₱2,000.00
CB-004	Information Management	Jake Maggay	2020	₱2,000.00
CB-005	Information Management	Jake Maggay	2020	₱2,000.00
CB-006	Information Management	Jake Maggay	2020	₱2,000.00
CB-007	Information Management	Jake Maggay	2020	₱2,000.00

In deleting a record, you should always use a unique identifier of a record in the condition or WHERE clause of the statement. In the example, we used the BookID of the book because it is the Primary Key the table and the only unique identifier of the record.

THANK YOU!